

Part D:

So, after researching this further and looking into the examples mentioned here is what I found. Firstly, Python (numpy) and R differ in terms of how they handle degrees of freedom when calculating standard deviation and variance. In Python it uses the population variance due to it using the default of $ddof = 0$ or delta degrees of freedom equal to 0. However, R uses sample variance which is better and less biased when calculating statistics by using $ddof = 1$ or that the samples is $n - 1$. Secondly, in Python nan values are skipped by default when calculating statistics (like the mean and sum) whereas R explicitly makes you decide how to deal with them before it will perform the calculation. Finally, R indexes based on 1 with inclusive intervals meaning that $[1,5]$ is 1 - 5 and includes the endpoints where as Python is $[1-5)$ but won't include 5 meaning that if you aren't expecting it, it can lead to an off by one error. Overall, most of the time R is either more accurate by default or requires you to be more specific before giving a result, whereas python gives a quick result but a result that may be less accurate due to making assumptions.